

Response to the Reviewer (Second Round)

Flatness and Passivity-Based Control with Adaptive Neural Compensation of a Direct-Drive PMSG Wind Energy Conversion System

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We thank the reviewer for the positive assessment and the remaining remarks, which we address below. Equation and section numbers refer to the revised manuscript. The paper keeps its six-page length; to make room, the full state model (a restatement of (4) and (8)–(10)) was replaced by a sentence and some text was condensed, without removing any result.

Reviewer: *The applicability of the neural stability theorem to the normalized adaptive update laws should be clarified.*

Response: We agree that the theorem of [15] does not apply directly. Its proof uses a Lyapunov function with $\text{tr}(\tilde{W}^T \Gamma^{-1} \tilde{W})$ and a constant gain; with $\Gamma_n(t) = \Gamma/(1 + \|\mathbf{z}_n\|^2)$ its derivative contains an extra term in $\dot{\Gamma}_n$. We removed the citation of that theorem and replaced it by a new, self-contained **Proposition 3** with its proof (Section V-C):

- (i) Because the learning signal is part of the network input, $\|\mathbf{r}\| \leq \|\mathbf{z}_n\|$, so $\Gamma_n \|\mathbf{r}\| \leq \Gamma/2$. With the σ -modification, this gives explicit bounds on $W_2, \mathbf{b}_2, W_1, \mathbf{b}_1$ for all t , whatever the tracking errors, e.g. $\|W_2(t)\| \leq \max(\|W_2(0)\|, \sqrt{10}\Gamma/(2\sigma))$.
- (ii) The compensation is bounded by its output saturation, and it enters the outer error equations (25)–(26) linearly. Proposition 2 shows that the nominal error system is exponentially stable, so it is locally input-to-state stable. Hence the tracking error is uniformly ultimately bounded, with a bound proportional to $\sup(|\Delta T_e - \Delta|/J + |\Delta P_g + \dot{y}_2^*|)$.

The text states the limitation explicitly: the result guarantees boundedness for any $\Gamma > 0$, but it does not prove that the network reduces the bound. That reduction is assessed numerically (Table IV), and the oscillations observed for $\Gamma \geq 20$ are consistent with bounds that grow with Γ/σ . As a consequence, the stability claims of the paper no longer depend on the unpublished reference [15].

Reviewer: *The approximation involving the derivative of the DC-link energy reference should be justified quantitatively.*

Response: Section V-A now quantifies this approximation:

- The reference y_2^* varies only through the filter energy, at most 300 J, compared with 14.4 kJ in the capacitor.
- The neglected term $\dot{y}_2^* = \frac{3}{2} L_f \dot{\mathbf{i}}_g^* \mathbf{i}_g^*$ enters the error equation (26) as an input. The L_1 norm of the impulse response of $s/(s^2 + k_{21}s + k_{22})$ is $2/(e \omega_p) = 12.3$ ms, so $|e_y| \leq 12.3 \text{ ms} \cdot \sup|\dot{y}_2^*|$.
- In the simulations, $|\dot{y}_2^*| \leq 2.7$ kW (0.14 % of P_n) outside the transients, i.e. $|e_y| < 34$ J, or less than 1.4 V on V_{dc} . It reaches 29 kW at start-up and 52 kW during the grid dip.
- We also simulated the law *with* $\dot{y}_2^*$ included. The ISE of V_{dc} improves by 3 %, but the dip deviation increases from 3.6 to 5.3 V, because $\dot{\mathbf{i}}_g^*$ must be obtained by numerical differentiation. This justifies omitting the term.

Proposition 2 now lists $\dot{y}_2^* = 0$ among its assumptions, and Proposition 3 includes $\dot{y}_2^*$ as an input.

Reviewer: *The scope of the differential flatness claim should be stated more precisely.*

Response: Section V-A now states that flatness is used only for the two outer subsystems, with the currents treated as inputs:

- the shaft equation (4), with input i_{sq} and flat output Ω ;
- the first-order energy balance of the DC link and filter, with input $P_g = \frac{3}{2} v_{gd} i_{gd}$ and flat output y_2 . The input is $P_g = P_{msc} - \frac{3}{2} R_f \|\mathbf{i}_g\|^2 - \dot{y}_2$, and V_{dc} follows from y_2 and the measured currents.

In both cases T_w (or its estimate), P_{msc} and i_{gq} are known exogenous signals. We state explicitly that no flatness property is claimed for the complete sixth-order model with the converter voltages as inputs. The coupling between the outer and inner loops is handled by Proposition 2.

Reviewer: *The bibliographic status of Reference [15] and the discussion of IEEE 519 compliance should be updated.*

Response: *Reference [15] is now cited as Mathematics, under review, 2026. It will be replaced by the full citation if it is accepted before the camera-ready version. As explained above, the paper no longer relies on [15] for any stability result; it is cited only as the origin of the compensation structure and of the wind profile.*

IEEE 519. In re-examining this point we found that the previous index was not the one defined by the standard. It summed every spectral bin up to 2 kHz and referred the result to the fundamental at partial load. Section VI-D has been rewritten as follows:

- As in IEEE 519-2022, the harmonics $h = 2, \dots, 50$ are referred to the rated current I_L (TDD). The TDD is 0.12, 0.07 and 0.13 % at 8, 9 and 10 m/s, and the largest individual harmonic is 0.11 % of I_L . Both are below the limits (5 % TDD; 0.3 % for $35 \leq h \leq 50$ in the strictest class).
- We also report what these indices do not cover. Since $f_{sw}/f_g = 43.2$, the switching components are interharmonics, at $f_{sw} \pm 2f_g$ and $f_{sw} \pm 4f_g$ (about 2.2 and 1.6 % of I_L each) and near $2f_{sw}$. In total they amount to about 4.6 % of I_L up to 5 kHz, nearly independent of the load.
- With a grid-synchronized carrier, these components would be harmonics above the 0.3 % limit. The paper therefore states that an *LCL* filter is required at this switching frequency, and lists it as future work.
- PI control gives the same switching components within 0.1 % of I_L : the ripple is set by the modulation and the filter, not by the control law.

We thank the reviewer again for the comments, which improved both the theoretical part and the validation of the paper.